



Analog IC Design

CMOS Technology

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Section 3 – Lab01

LPF and MOSFET characteristics

Lab (1) PART 1: Low Pass Filter Simulation (LPF)

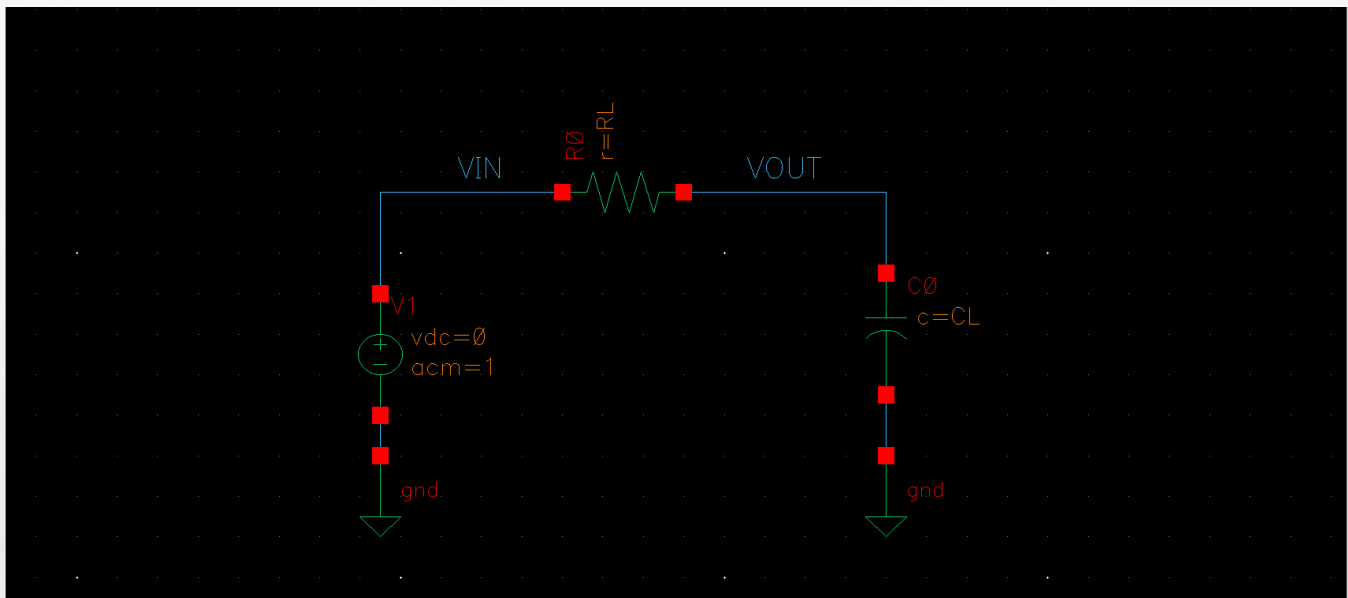
1. Transient Analysis

1) $R = 1k\Omega$ and $0.5ns$ time constant.

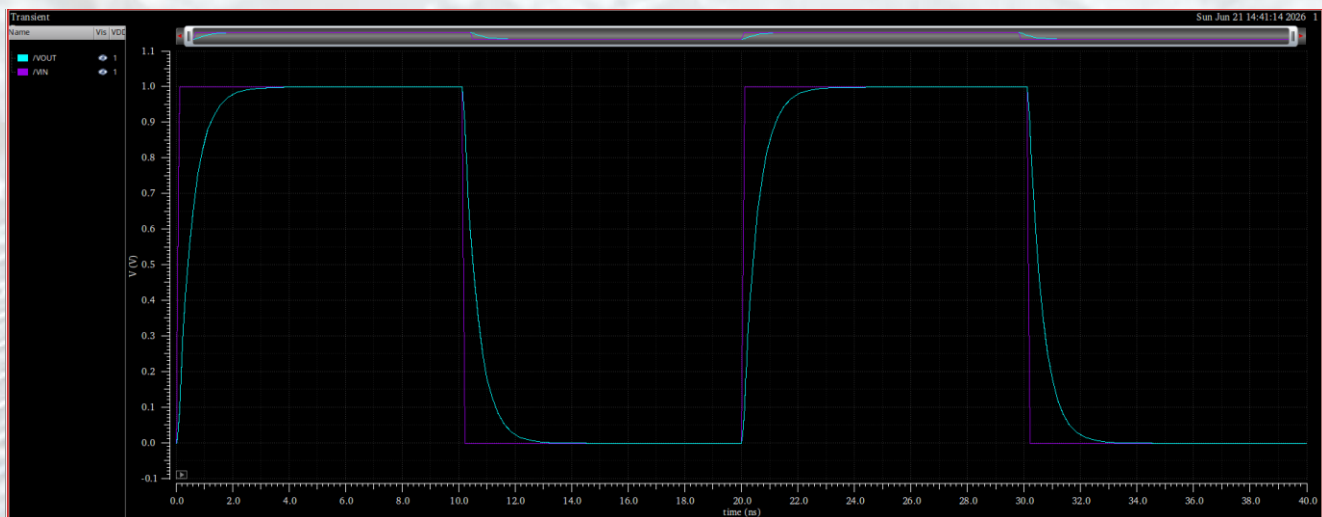
$$\tau = RC \rightarrow C = \frac{\tau}{R} = \frac{0.5ns}{1k\Omega} = 0.5pF$$

2)

Applied a square wave input with $T_{high} = \text{Pulse Width} = 10ns$, $T_{CLK} = \text{Period} = 20ns$, and $T_{rise} = T_{fall} = 100ps$, $V_{DD} = 1v$.



3)



V_{out} , and V_{in} waveforms

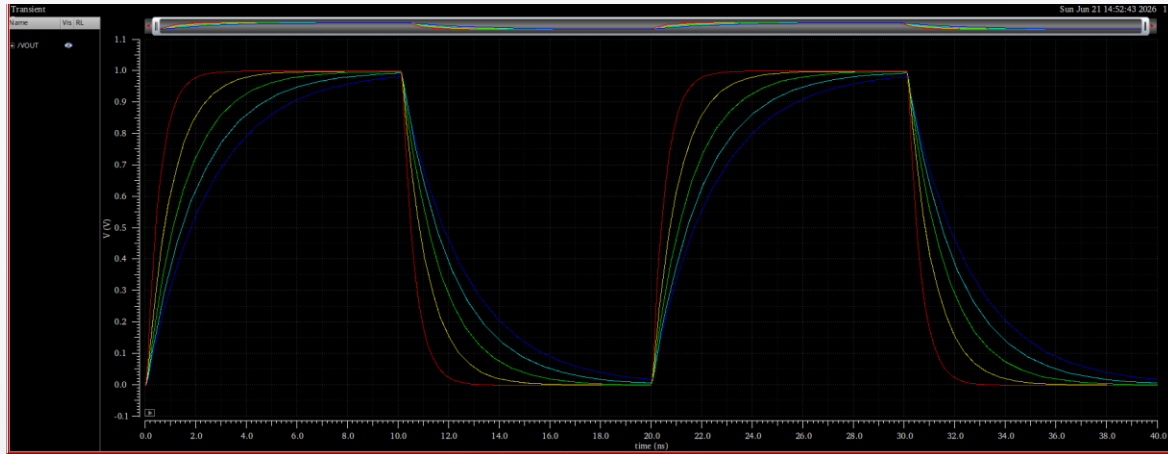
4) from simulator:

Rise time = Fall time = 1.104 ns

5) Step response (10% to 90%): rise time = fall time $\approx 2.2 \tau = 2.2 * 0.5 \text{ ns} = 1.1 \text{ ns}$

Parameter	Analytical	Simulation	Error
Rise time	1.1 ns	1.104 ns	0.004 ns
Fall time	1.1 ns	1.104 ns	0.004 ns

6) sweep for $R = 1:1:5 \text{ k}\Omega$.



The change in V_{out} with R

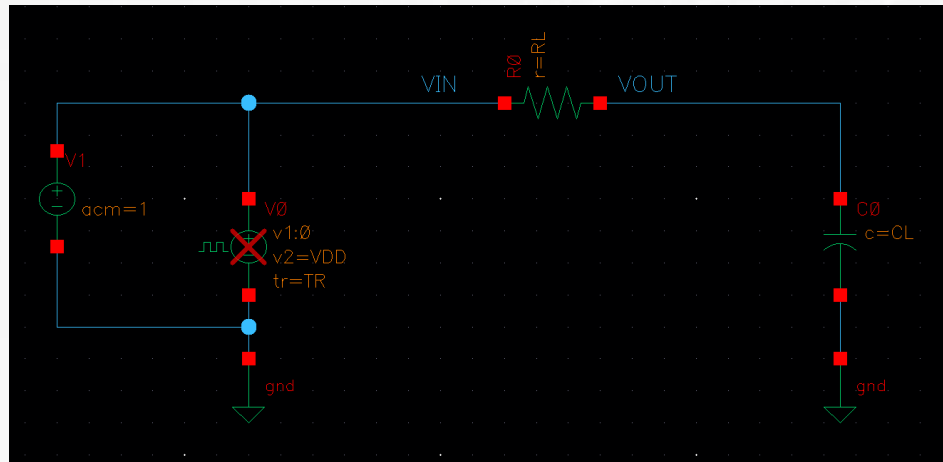
Comment:

As R increases:

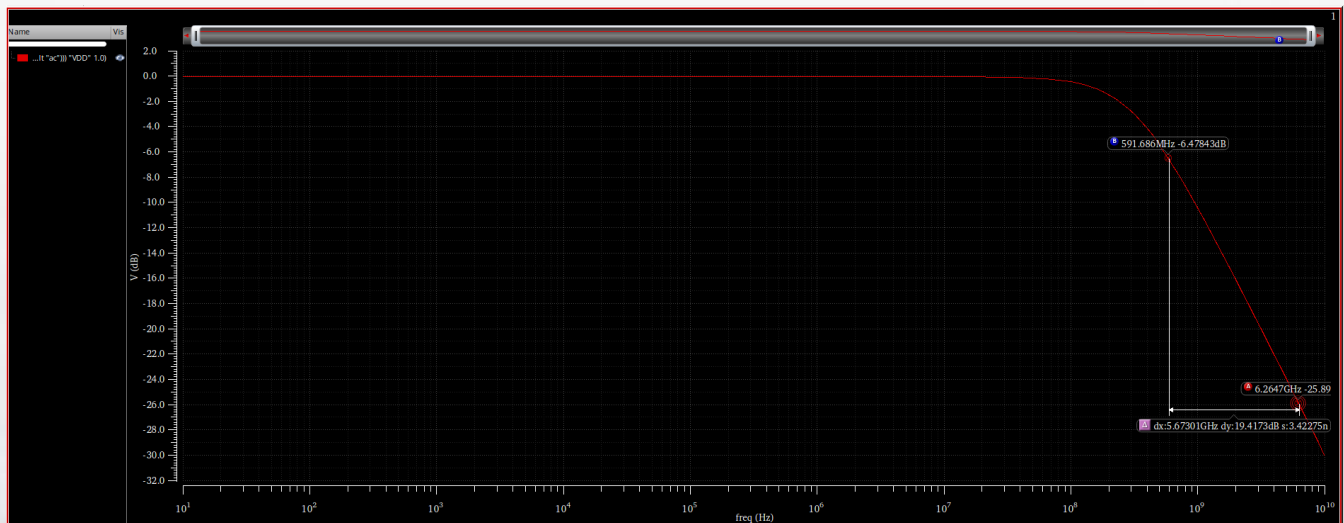
The time constant ($\tau = RC$) also increases, and the output voltage will takes **more** time to charge or discharge, so **both** rise and fall time increases (slower output).

“The filter becomes more smoothing”

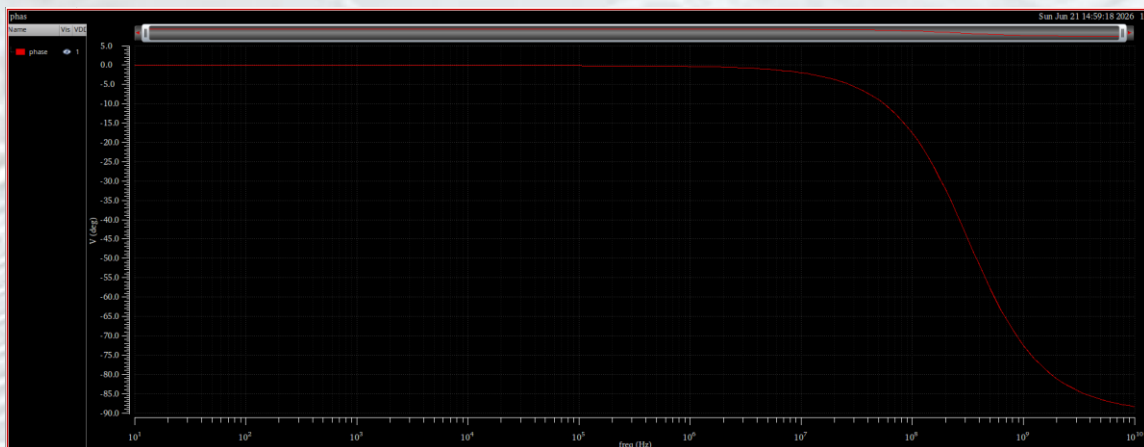
2. AC Analysis



1) Bode Plot.



Bode plot: magnitude ($dx \approx 1$ decade, $dy = 19.4173\text{dB} \approx 20\text{dB}$)



Bode plot: phase

2) DC gain and 3dB bandwidth from simulator:

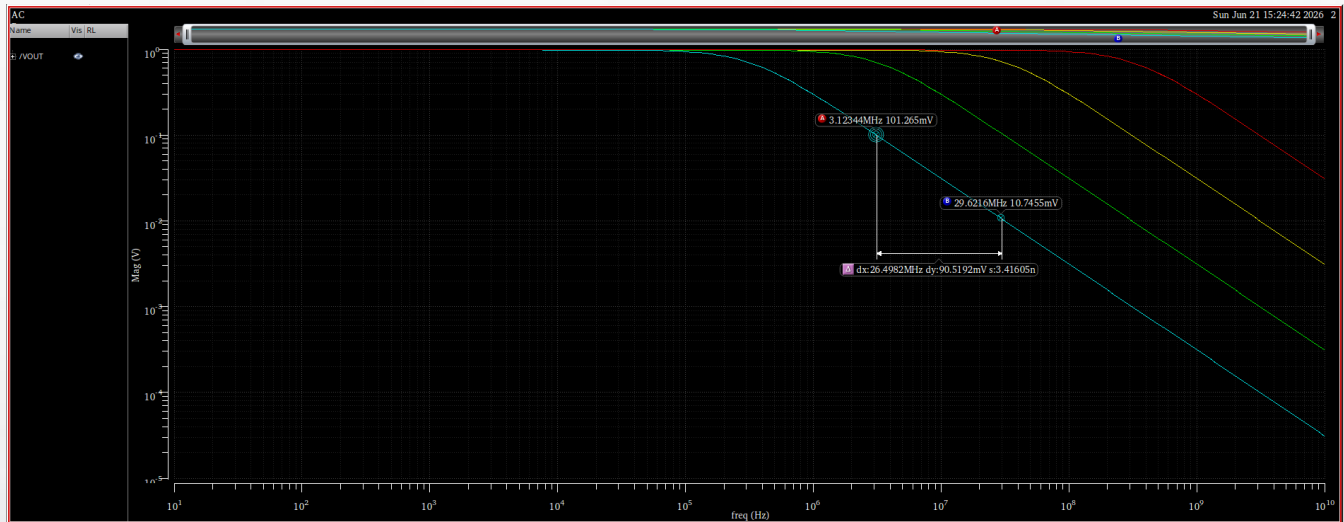
DC gain = 1 , 3dB bandwidth = 317.6MHz

3) DC gain $= |H| = \frac{1}{\sqrt{1+(\omega RC)^2}}, f = 0 \rightarrow \omega = 0 \rightarrow |H| = 1$

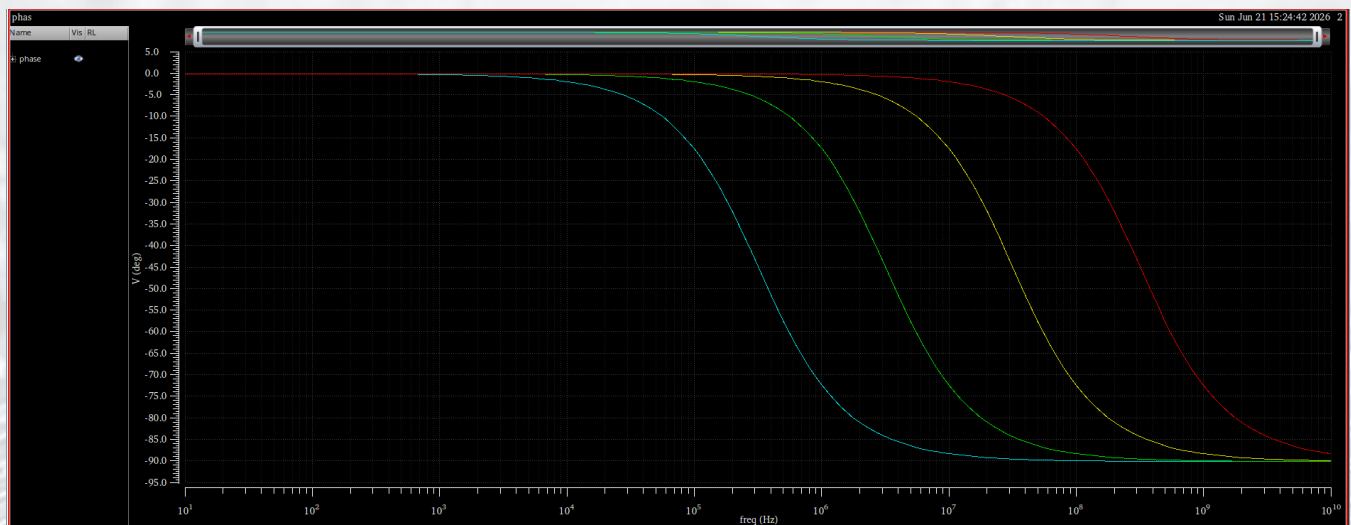
3dB bandwidth $= f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1k\Omega)(0.5pF)} \approx 318.3MHz$

Parameter	Analytical	Simulation	Error
DC gain	1	1	0
3dB bandwidth	318.3MHz	317.6MHz	0.7MHz

4) sweep for $R = 1, 10, 100, 1000k\Omega$.



Change in the magnitude with BW (log scale)



Change in the phases with BW

Comment:

As R increases:

The cutoff frequency **shifts** to left, **and** the Bandwidth decreases.

The magnitude roll-off remains ($\approx -20dB/decade$).

3. Pole-Zero Analysis

1)

$$H(s) = \frac{1}{1+sRC} \rightarrow \text{pole: } s = -\frac{1}{RC} = -\frac{1}{(1\text{k}\Omega)(0.5\text{pF})} = -2 \times 10^9 \text{ rad/s}$$

2)

PZ Analysis 'pz'

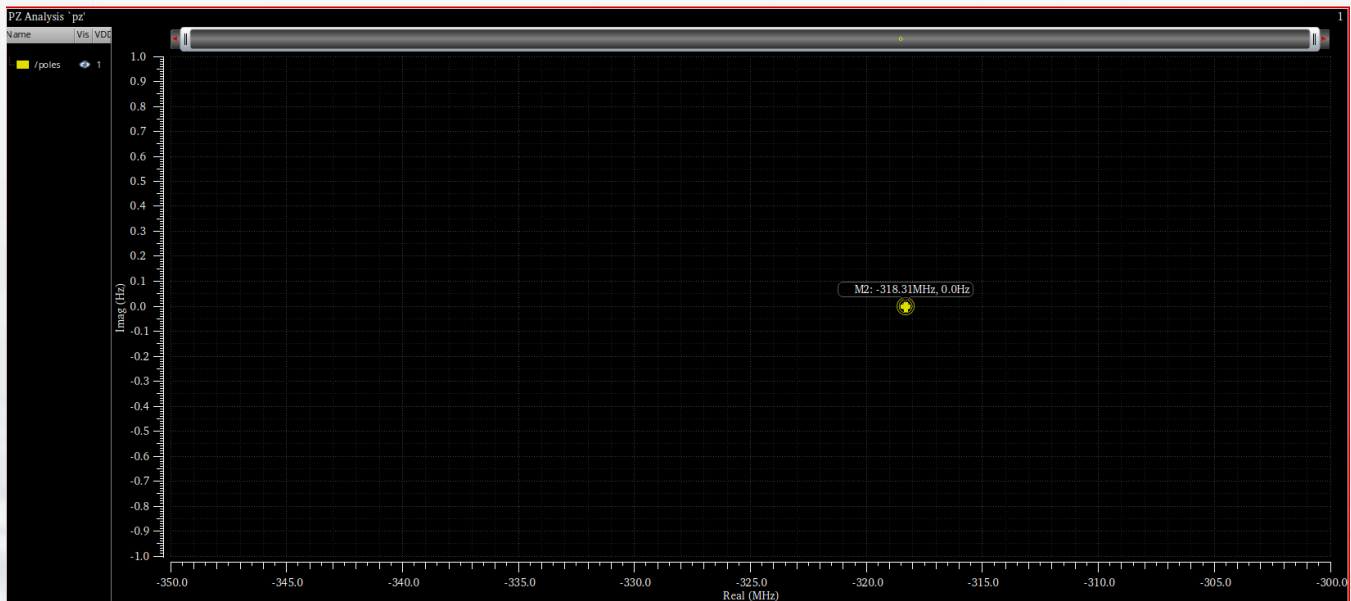
DC simulation time: CPU = 189 us, elapsed = 190.02 us.

DC simulation time: CPU = 31 us, elapsed = 30.9944 us.

Poles (Hz)

	Real	Imaginary	Qfactor
1	-3.18310e+08	0.00000e+00	5.00000e-01

No zero is found

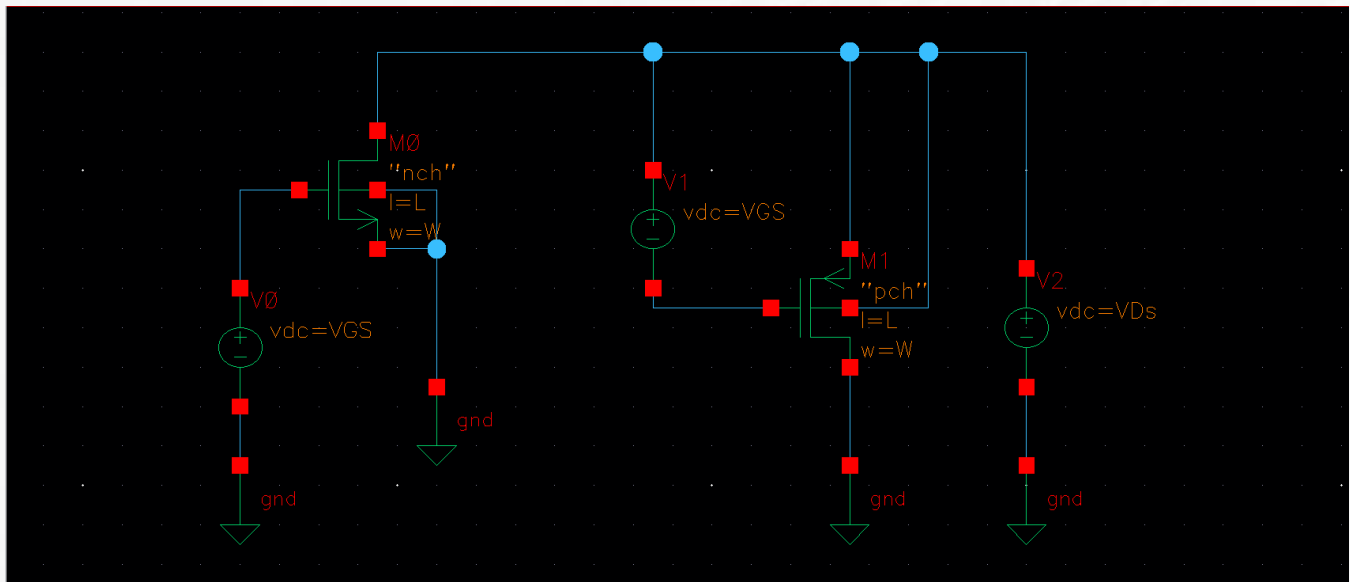


P-Z graph (one pole on the real axis at $f_p = -318.31\text{MHz}$ and no zeros)

The pole frequency $|f_p| = \frac{|s|}{2\pi} = \frac{2 \times 10^9 \text{ rad/s}}{2\pi} = 318.3\text{MHz}$

The calculated BW from the AC analysis = 318.3MHz (same as the f_p calculated from P-Z analysis).

Lab (1) Part 2: MOSFET Characteristics



1. I_D vs V_{GS}

1)

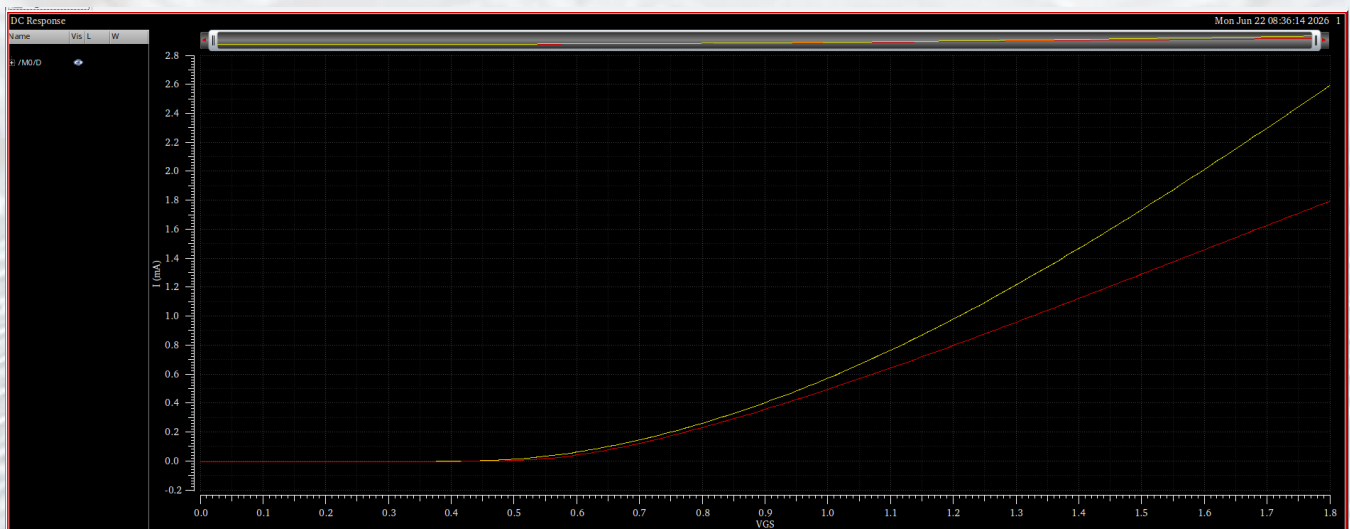
The technology used: 180nm $\rightarrow V_{DD} = 1.8V$

$V_{DS} = V_{DD} = 1.8V, V_{GS} = 0:1mV:1.8V$

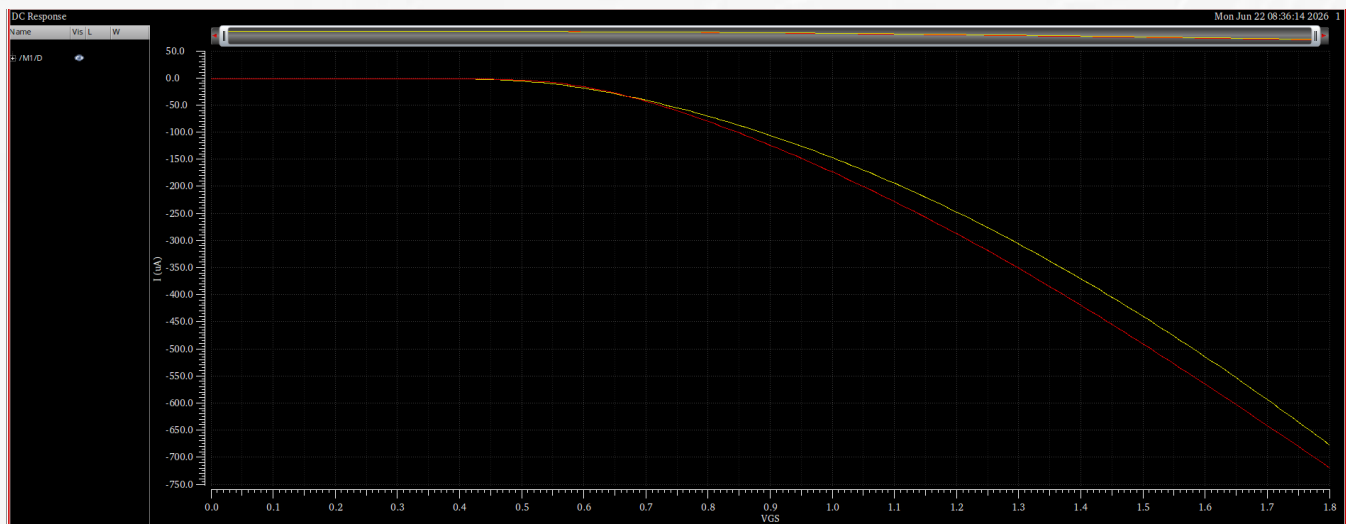
$W/L = 15 \rightarrow W = 15L$

Short channel test: $W = 3\mu m, L = 200nm$

Long channel test: $W = 30\mu m, L = 2\mu m$



NMOS characteristics: ■ long channel, ■ short channel



PMOS characteristics: ■ Long Channel, ■ Short channel

2) the differences between short channel and long channel results

Which one has higher current? Why?

Is the relation linear or quadratic? Why?

For **NMOS** technology:

The long channel has higher current (for high applied voltage).

The electric field ($E \propto \frac{1}{L}$ at const. applied voltage) in a short channel is stronger than the long channel, so it forces the electrons to reach their maximum speed **faster** at lower applied voltage than the long channel, and the current can't increase as it does in the long channel.

For **long** channel:

The relation is quadratic and follows the square law behavior $I_D \propto (V_{GS} - V_{TH})^2$

For **short** channel:

The relation starts with square law behavior at low electric field.

Then gradually transitions to a weaker dependance on it (quasi-linear) at high fields **due** to velocity saturation.

For **PMOS** technology:

The short channel has a little bit higher current (for high applied voltage).

Since PMOS carriers have lower mobility than NMOS, velocity saturation is generally **less** dominant. Instead, short channel effects such as threshold voltage roll-off become more **significant**, reducing the effective threshold voltage and allowing the short channel device to turn on more strongly and conduct **higher** current.

Both long and short channels show approximately **quadratic** behavior.

The long channel device follows the square law more ideally, while the short channel device shows slight deviations **due** to short channel effects.

3) the differences between NMOS and PMOS.

Which one has higher current? Why?

NMOS has higher current since its carriers (electrons) have higher mobility than the PMOS carriers (holes).

the ratio between NMOS and PMOS currents at $V_{GS} = V_{DD} = 1.8\text{V}$

for long channel: $I_N \approx 2.6\text{mA}$, $I_P \approx 0.675\text{mA}$ $\rightarrow \frac{I_N}{I_P} = \frac{2.6}{0.675} = 3.85$

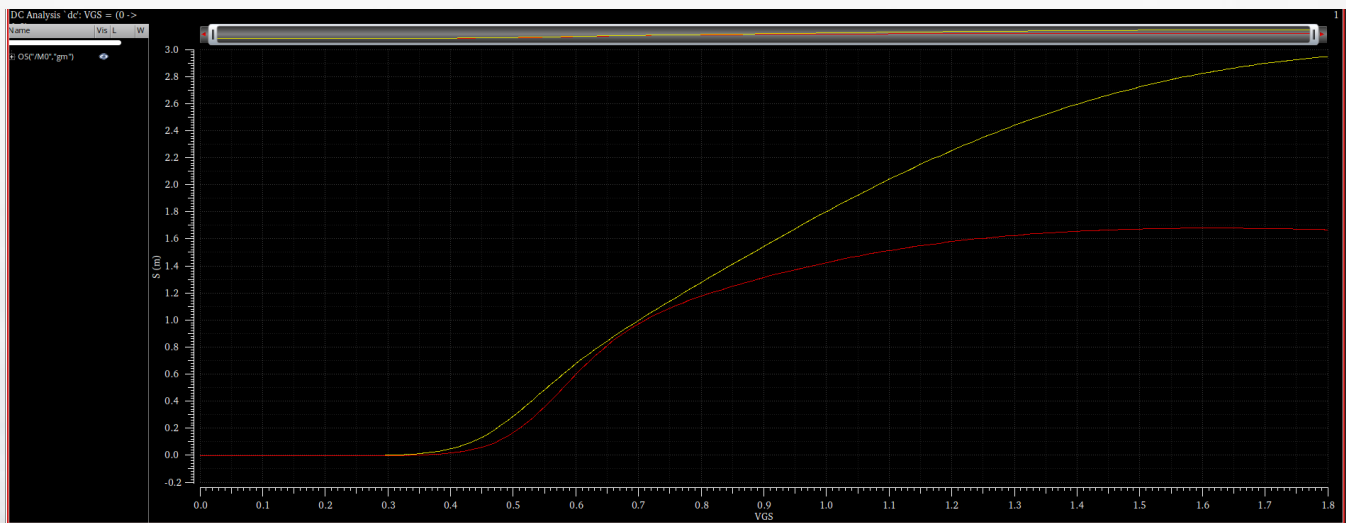
for short channel: $I_N \approx 1.8\text{mA}$, $I_P \approx 0.725\text{mA}$ $\rightarrow \frac{I_N}{I_P} = \frac{1.8}{0.725} = 2.48$

Which one is more affected by short channel effects?

NMOS

2. g_m vs V_{GS}

1)



g_m vs V_{GS} for NMOS: ■ long channel, ■ short channel

2) the differences between short channel and long channel results.

Does g_m increase linearly? Why?

In long channel:

g_m Increase linearly at **low** voltage, **due** to the linear dependence between g_m and V_{GS} at low voltage. But then the increasing rate slows down

In short channel:

g_m increases fast “exponentially” at low voltage then saturate at high voltage, since the electronics reach a velocity saturation.

Does g_m saturate? Why?

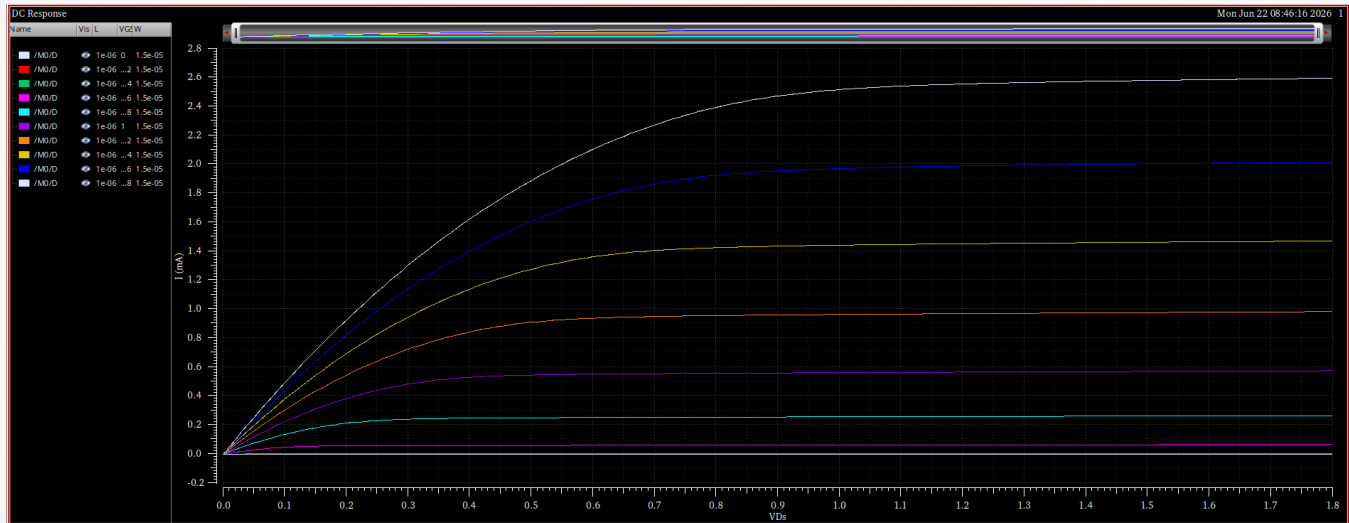
g_m will saturate for short channel at high fields.

Since the electron's velocities will saturate and the relation between I_D and V_{GS} becomes quasi-

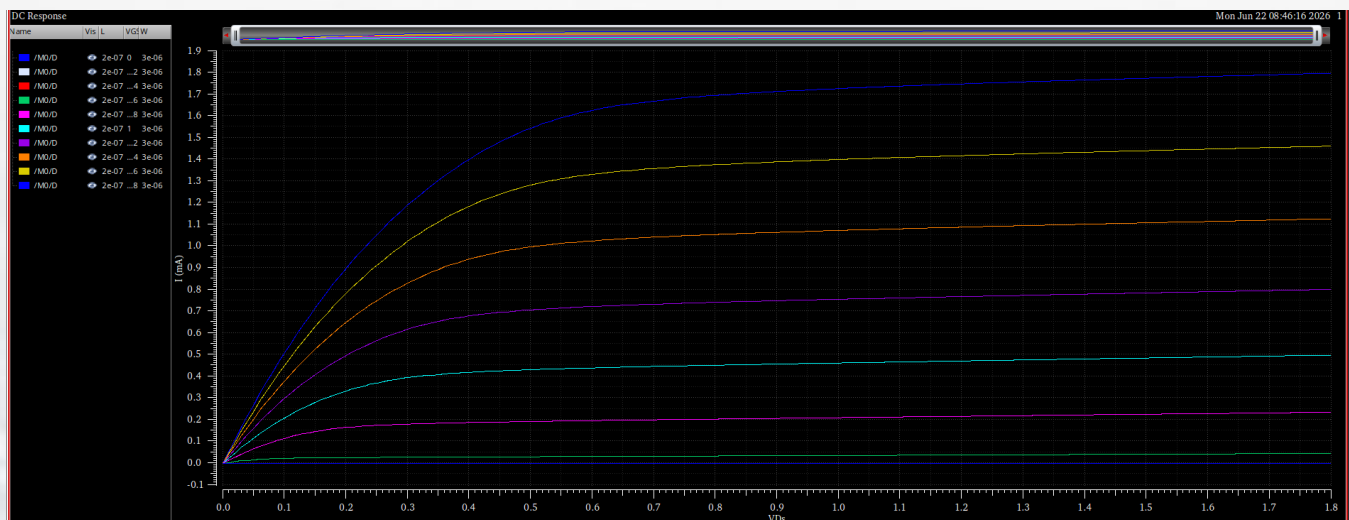
linear. $g_m = \frac{\partial I_D}{\partial V_{GS}} \approx \text{constant}$.

3. I_D vs V_{DS}

1)



Long channel NMOS $I_D - V_{DS}$



Short channel NMOS $I_D - V_{DS}$

2) the differences between short channel and long channel results.

Which one has higher current? Why?

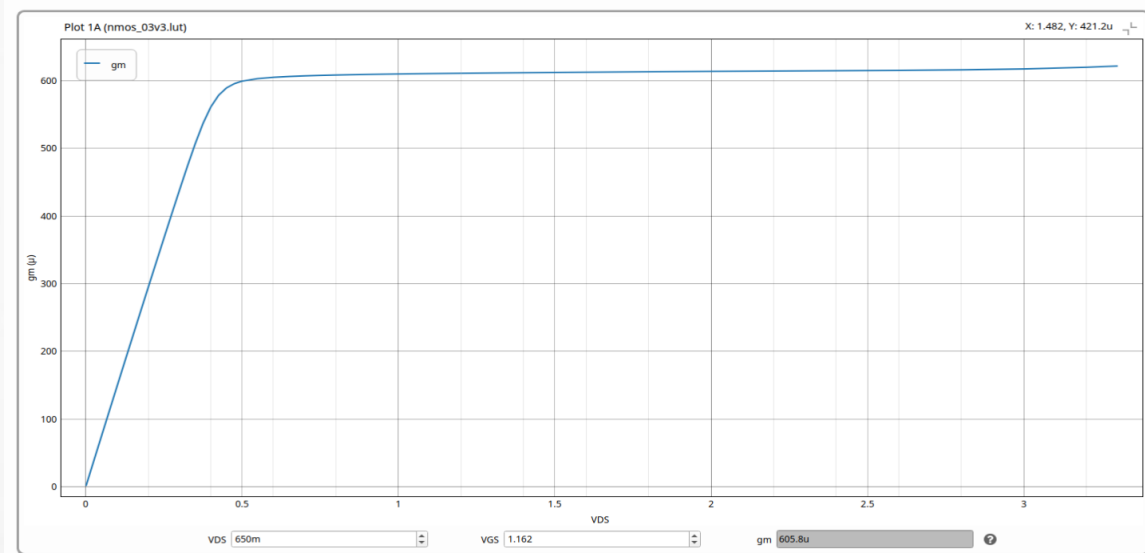
The long channel has higher current at the same applied voltage, **due** to the velocity saturation, since ($E \propto \frac{1}{L}$ at const. applied voltage) in short channel will reach its velocity saturation at voltage **lower** than the long channel.

Which one has higher slope in the saturation region? Why?

The short channel has a higher slope, since the channel pinched off close to the drain after a certain V_{DS} , as V_{DS} increase the pinch off point moves toward the source **decreasing** the effective length. Since ($I_D \propto \frac{1}{L}$), as the effective length decreases the drain current increases. For short channel the decreasing in the effective length is **large** relative to the length so it represents

a higher slope. For long channel the decreasing in the effective length is **small** relative to the length so it seems like a horizontal line.

g_m and r_o



g_m vs V_{DS}

Comment:

In the first part of the curve, is the relation linear? Why?

Yes, since the current in this region is given by the quadratic relation: $I_D = k'_n(V_{ov}V_{DS} - \frac{V_{DS}^2}{2})$,

$$V_{ov} \propto V_{GS}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = k'_n V_{DS} \rightarrow \text{linear relation with } V_{DS}.$$

Does g_m saturate? Why?

Yes, when $V_{DS} > (V_{ov} = 0.5v)$ the current will be given by the relation: $I_D = \frac{k'_n}{2} V_{ov}^2$, $g_m =$

$\frac{\partial I_D}{\partial V_{GS}} = k'_n V_{ov}$ doesn't depend on V_{DS} in ideal case, but actually there will be small dependence

on V_{DS} , $g_m = k'_n V_{ov}(1 + \lambda V_{DS})$

How much is V_{DS} when the curve saturates?

Once $V_{DS} \geq V_{ov} = 0.5v$ the curve will saturate.

Where do you want to operate the transistor for analog amplifier applications? Why

The best region to operate the transistor as an amplifier is the **saturation** region, since in this region the transistor acts as VCCS which can get **large** and **linear** gain.